

ccint — Progress Brief

Canada Cyber Social Intelligence: why it is built this way, what was built, and where it stands

23 September 2026

The question, and what changed about it

The project began as a longitudinal study: *how does Canada-related cybersecurity discussion change over time on social media?* A collector runs continuously, posts are labelled under versioned schemes, and change is tested statistically.

The first full analysis returned a negative result, and returned it with enough rigour to be worth something. The largest topic movement in the 30-day window, `data_breach` at +22%, has $p_{\text{FWER}} = 0.734$ under an author-level circular-shift null that preserves each author's volume, topic mix and burstiness while destroying calendar alignment. Empirical power analysis — measured by author-pool replication rather than assumed — puts the requirement at roughly $8\times$ the current volume.

That result is only interesting if one knows what produced it. Three explanations were open — the world, the relevance gate, or the collection boundary — and nothing in the system could separate them, because every figure it produced came from labels it had produced itself. Most of the work since has been building measurements that can, and the answer was not the expected one: **the shortfall is not in the statistics and not mainly in the labelling — the platform carries almost no human discussion of these events at all.**

Why the system is built this way

Four commitments shaped every design decision. Each exists to prevent a specific failure that would be invisible after the fact.

One platform first, and say so. An open, documented API with usable historical search and terms permitting research collection; other platforms were excluded on access grounds rather than merit. This is a convenience sample, stated rather than hidden, and measuring what such a sample can show became part of the job.

Labels are versioned and never overwritten. Annotation rows are keyed by (post, annotator version) and only ever inserted; eight annotator versions now coexist over the same corpus and diff row by row. Otherwise every improvement to the annotator silently rewrites history, and a shift in topic distribution can never be attributed to the world rather than to the method.

Raw data is kept verbatim; every number carries provenance. Payloads are stored unmodified, and every run and model label records what produced it, including the SHA-256 of the prompt text. That last part was added after noticing that a model annotator's output is entirely determined by its prompt: "labels are never overwritten" had preserved the labels while losing the conditions that made them.

System failure must be distinguishable from social change. Every collection path records mode, status, window and counts. The principle held under test: a token outage appears as 43 `partial` runs, not as a drop in Canadian cyber discussion.

Underlying all four: **measure before improving.** Repeatedly the intuitive next step was aimed at the wrong quantity — tuning the trend score would have been wasted, relaxing the relevance rules helps but not for the population that seemed to need it, and the largest prompt improvement in the project introduced a regression the evaluation set certified as correct.

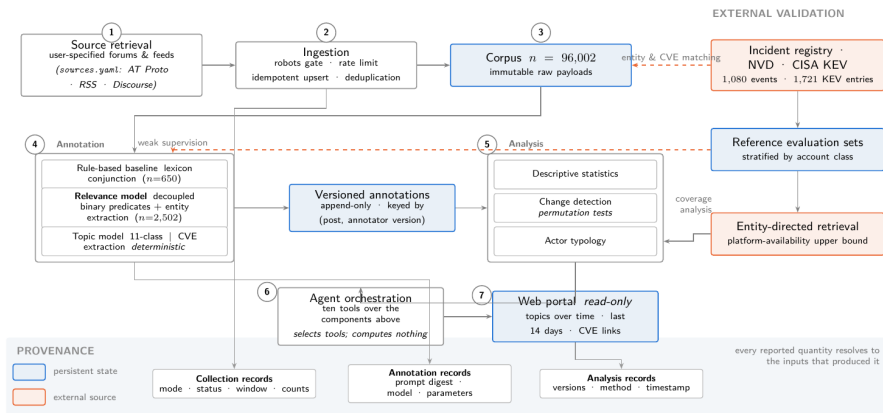


Figure 1: Processing pipeline. Stages 1–3 acquire and store the corpus, 4 annotates it under coexisting versioned schemes, 5 analyses it, and 6–7 expose it through a tool-calling agent and a read-only portal. The external-validation column supplies labels no component of the pipeline produced.

What was built

Collection runs in two modes — a day-by-day backfill and an hourly incremental — with idempotent ingestion, stale-run reaping and client-timestamp repair; the corpus stands at 96,002 posts across three sources. Annotation proceeded in generations, all retained. Analysis covers descriptive statistics, permutation testing against two null models, behavioural broadcast detection with a collection-lag control, and a two-axis actor typology separating measured audience from content-judged function. Later phases added external validation, then user-specified sources, deterministic CVE linking, an agent orchestration layer and a read-only portal.

Sources are configuration, not code. `config/sources.yaml` names the forums and feeds to read; two generic collectors (RSS/Atom, the Discourse public JSON API) cover most of what a security community exposes, both consult `robots.txt` and refuse a source they cannot read — a blocked collector produces a gap, and a gap is indistinguishable from a fall in discussion — and each source must declare why automated access is believed permissible.

Where it stands

The collection boundary was never measured

The 34 search terms came from a specification naming four examples followed by “etc.”; the remaining 26 were supplied during implementation from domain intuition, never derived from data and never revised. Reconstructed offline over the stored corpus, per-term productivity is close to degenerate: `cybersecurity` returned 22,498 posts and 200 relevant ones, `CVE 12,502` and `three`; eleven terms returned no relevant post at all. `cybersecurity` alone accounts for 30.8% of relevant posts, so the corpus time series is to first order one search term’s, and nothing monitored it. None of the 34 terms names Canada, so any Canadian cyber post not phrased with one of them is unreachable by construction.

Replacing the relevance gate

The old decision was a conjunction of two narrow lexicons, so recall was bounded by their intersection: 650 of 91,551 posts passed and the 89,714 rejected had never been examined by anything semantic. The replacement poses two *independent* predicates instead — a single combined prompt makes a 4B model collapse the conjunction onto the side it knows better, admitting global threat-intelligence noise at 6.8% against a true Canadian rate of 1.1%.

Relevance gate	recall @ registry	specificity @ registry	corpus relevant
rules_v2 — keyword conjunction	46.5%	99.8%	650
llm_v2c — two questions + validation	68.3%	99.4%	2,348

Of the 1,799 posts newly admitted, 84.0% had been rejected on the Canadian side of the conjunction. Two results here are method rather than number. Entity offsets are computed with `str.find` rather than requested from the model, on the narrow grounds that small models miscount them — which turned into a free hallucination detector, since an entity that cannot be located in the post was invented, and nearly half of extracted `place` entities failed it. And the residual error is neither prompt nor capability: the posts still missed carry no country marker at all, so closing it needs a gazetteer — which is the same registry currently used to measure the gap.

The external reference, and the central result

Until this phase every precision and recall figure traced to a single author who wrote the rules, the prompts and the evaluation set; a reported $\kappa = 0.870$ measured agreement under those conditions, not independence. A public ransomware victim registry (1,080 events, 1,943 entities) supplies labels no part of the system produced. For every Canadian event in the window the victim’s name was then searched directly with no cybersecurity term attached, each hit passing the cyber judgement before counting:

Canadian ransomware events, 22 Aug – 22 Sep 2026	Events	Share
Listed in the registry	27	—
At least one cyber-related post exists on the platform	26	96%
At least one such post was collected by ccint	26	96%
At least one came from a non-feed account	2	7%

The 144 relevant posts come from 15 accounts. Twelve are automated threat-intelligence feeds; the three that are not are Canadian news outlets contributing five posts between them, all about one event. Individual users contributed nothing, and 94.9% of the posts have no likes at all.

What looks like coverage is syndication. The collector is not the bottleneck — it captured 26 of the 26 events the platform carried. No ransomware trend is detectable because there is almost no ransomware *discussion* to detect, which also reframes the power analysis: 8× of this corpus is 8× as much syndication. The one exception is instructive — a Quebec construction regulator listed by Qilin drew real bilingual media coverage, and the single post the collector missed carries no term from the 34-term list, because *protection préventive* is in no lexicon.

Linking discussion to vulnerabilities

CVE identifiers are extracted by regular expression, not by a model: the format is fixed by MITRE, which makes this the one table in the system whose trustworthiness does not depend on annotation quality. They are enriched from two independent external sources — NVD for theoretical severity, the CISA Known Exploited Vulnerabilities catalogue for observed exploitation — kept in separate columns and never combined into a risk score, because one is a judgement about how bad a flaw could be and the other an observation that someone is using it.

Across the corpus, 10,820 posts mention 5,404 distinct identifiers. Only 147 — 2.7% — are in the KEV catalogue, yet they attract 19.3% of all mentions, a sevenfold concentration, and 21 of the 25 most-discussed identifiers are among them.

Attention tracks exploitation, not severity. This is the first result in the project where attention aligns with an external ground truth rather than mirroring a feed, and the first directly usable one: ranking vulnerabilities by how much they are discussed recovers, unprompted, most of the set CISA has independently confirmed as exploited.

Making the system answerable

The most recent phase adds no analysis. It turns the existing components into ten tools, lets a local model call them, and puts the results behind a read-only portal with a source selector, topic trajectories, the last fourteen days, the CVE table and collection health side by side. The tools compute nothing of their own: `get_actor_composition` forwards to the same broadcast detector the analysis uses, and a test fails if a second copy of that threshold appears — two implementations of one definition drift apart, and nobody notices until a number is wrong in public.

The agent's constraints are the project's own measurements, written into the system prompt: volume is not attention, so the feed share must be reported beside it; this corpus describes activity in the monitored sources, not Canadian public opinion; movement is not a trend, since permutation testing found none distinguishable from noise; and every claim must name its source, window and counts. Asked whether ransomware activity is organic or feed-driven, the agent chose three tools across four turns and answered correctly — with the feed share, the scope limit, and a refusal to call the change a trend.

Building this layer surfaced two defects of a kind that only appears when numbers go on a screen for someone else. Topic queries folded posts with no topic row into the residual `other` bucket — the same error already ruled out in the actor typology, where `unclassified` is never a negative; labelling them (`unlabelled`) exposed 962 recent posts with no topic label at all. And the portal's topic-version constant still pointed at a superseded annotator.

What the project is, and what comes next

Trend analysis of Canadian cyber discourse is not available at this volume for this event class, and adding data will not make it so. What is available, and appears unreported in the surveyed literature, is a sensor that states its own blind spots numerically. Published systems report a numerator (84 of 151 events detected, in the strongest comparable work); none reports how many events had no platform presence at all, because establishing that requires the per-event entity-directed retrieval used here — affordable only when the event count is small. This project's greatest liability, a corpus of a few thousand relevant posts, is what makes the denominator measurable. Immediate priorities, in order:

1. **Widen the registry beyond ransomware** — advisories, KEV, Canadian media. Every figure above is conditioned on a source that sees only leak-site victims: the narrowest limitation in the work.
2. **Feed the registry back into the relevance decision**, since the residual failure is missing knowledge of exactly the names it lists.
3. **Add a Canada-side query stream**, disjunctive rather than filtering, to reach the population the registry cannot see.
4. **Have an independent annotator label the 200-post sample** already drawn and left blank — the one step that cannot be automated.
5. **Let the new sources accumulate**, and add a paginated one: RSS cannot be backfilled, so the two feeds so far gave 45 posts.

Corpus 96,002 posts, three sources, 22 Aug – 23 Sep 2026; gate evaluation measured 22 Sep against 91,551 posts; eight coexisting annotator versions; 268 tests passing. Methodology and limitations in the full report.